🔑 **Answer Key — Teacher Copy · Lesson 9-4**

MSTAR PRACTICE WORKSHEET

Lesson 9-4: Apply Two-Variable Relationships to Solve Problems

Grade 6 Mathematics · Standard 6.AT.11

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 SELECTED RESPONSE — CORRECT: C

The assembly line produces 2.5 cars per hour. How many cars does it produce in one 8-hour shift?

A. 10.5 cars

B. 16 cars

C. 20 cars

D. 25 cars

Substitute 8 hours: 2.5 × 8 = 20 cars per shift.

Question 2 SELECTED RESPONSE — CORRECT: A

Let h = hours of operation and c = cars produced. Which equation represents the assembly line?

A. c = 2.5h

B. c = 2.5 + h

C. h = 2.5c

D. c = h ÷ 2.5

Every hour the plant produces 2.5 cars, so the total is 2.5 times the hours: c = 2.5h. Check with the table: 8 hours gives 20 cars.

Question 3 SELECTED RESPONSE — CORRECT: D

Using the table (1 hour, 2.5 cars; 8 hours, 20 cars; 16 hours, 40 cars), how many cars are produced in 24 hours?

A 26.5 cars

B 40 cars

C 50 cars

D 60 cars

24 hours at 2.5 cars per hour is 2.5 × 24 = 60 cars — three full shifts of 20.

Question 4 SELECTED RESPONSE — CORRECT: B

Last year the student council washed 34 cars and raised $408. How much did they charge per car?

A $10

B $12

C $34

D $374

Divide the amount raised by the number of cars: 408 ÷ 34 = 12 dollars per car.

Question 5 SELECTED RESPONSE — CORRECT: D

At $12 per car, how much does the student council raise by washing 20 cars?

A $32

B $120

C $200

D $240

Substitute 20 cars into t = 12n: 12 × 20 = 240 dollars.

Question 6 SELECTED RESPONSE — CORRECT: B

The price is set at $12 per car. Which is the DEPENDENT variable at the car wash?

A. The number of cars washed

B. The total amount of money raised

C. The price of $12 per car

D. The number of students on the council

The total raised responds to how many cars get washed — money raised depends on cars washed.

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7 TWO-PART QUESTION

Part A — correct answer: A

A bottling machine fills 45 bottles each minute. The relationship is b = 45m, where m is the number of minutes the machine runs and b is the number of bottles filled. The crew needs 1,800 bottles. How many minutes must the machine run?

A. 40 minutes

B. 45 minutes

C. 1,755 minutes

D. 81,000 minutes

Substitute 1,800 for b to get 45m = 1,800, then divide both sides by 45: m = 1,800 ÷ 45 = 40 minutes. Checking the answer, 45 × 40 = 1,800 bottles.

Part B — correct answer: C

Which reasoning justifies the number of minutes chosen in Part A?

A. The machine fills 45 bottles in a minute, so 1,800 bottles take 45 minutes.

B. Multiply instead: 1,800 × 45 = 81,000, because a rate is always multiplied by the total.

C. Substitute 1,800 for b to get 45m = 1,800, then divide both sides by 45, since 45 is multiplied by m.

D. Subtract the rate: 1,800 − 45 = 1,755, because the machine already filled 45 bottles.

In b = 45m the 45 is multiplied by m, so dividing both sides by 45 undoes the multiplication and gives m = 40. Multiplying 1,800 by 45 gives 81,000, far more minutes than the whole order needs, and subtracting treats the rate as a one-time amount instead of a rate for every minute.

Question 8 WRITTEN RESPONSE · 2 POINTS

Type II Reasoning: Rounding Down on an At Least Problem

The paint booth coats 6 car doors every hour, so d = 6h, where h is the hours the booth runs and d is the doors coated. Marisol needs 100 doors coated. She writes 6h = 100, divides to get 100 ÷ 6 = 16.66..., says the booth needs about 16 hours, and schedules 16 hours.

Explain Marisol's misconception and give the correct number of whole hours the booth must run.

Model answer: Marisol rounded down. Substituting h = 16 into d = 6h gives 6 × 16 = 96 doors, which is fewer than the 100 doors needed. Because the booth must coat at least 100 doors, the leftover part of an hour still has to be run, so the answer rounds up to 17 hours. Substituting h = 17 gives 6 × 17 = 102 doors, which meets the goal.

2 points	Student explains that 16 hours coats only 96 doors, so the leftover part of an hour cannot be dropped when the goal must still be met, and gives 17 hours.
1 point	Student gives 17 hours but does not explain why rounding down leaves the goal unmet.
0 points	Student agrees with Marisol or gives an unrelated response.

Answer Key. Score written responses with the rubric and guidance above; award partial credit per the 1-point rows.